AI SYSTEM SPECIFICATION

NAME: Salal Shaikh

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SUBJECT: ARTIFICIAL INTELLIGENCE

ASSIGNMENT: AI SYSTEM SPECIFICATION

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Chosen System: A Fraud Detection System for Digital Payments and Banking Transactions

Task 1: PEAS Specification

Performance Measures

The system evaluates operational success by minimizing financial exposure while maintaining seamless banking availability. Core performance criteria include:

- True Positive Rate (percentage of actual fraudulent activities successfully intercepted)
- Precision score (ratio of genuine fraud cases out of all transactions flagged by the model)
- Real-time verification speed (computational latency in evaluating authorization requests)
- Total capital protected (monetary value preserved from illicit extractions)
- Customer friction metric (volume of valid purchases disrupted by false alerts)

Environment

The system is deployed directly inside a complex decentralized processing framework connected to international banking pipelines.

The environment consists of primary account users, fraudulent operators, merchant servers, credit card clearinghouses, and highly variable traffic surges. Network bandwidth speeds fluctuate significantly across geographical endpoints, and incomplete payload packets or encrypted data anomalies can impact raw telemetry extraction. The layout is structured around institutional regulations but behaves unpredictably due to shifting consumer behavior and global attack vectors.

Actuators

The system exerts control over its environment utilizing the following physical and logical interfaces:

- Automated routing logic to clear, flag, or instantly reject pending wire transfers
- Dynamic verification step-up mechanisms to deploy real-time OTP or security challenges
- Cryptographic authorization holds to isolate suspicious checking or savings assets
- Security operational console interfaces to push risk-score profiles to fraud investigation teams

Sensors

The underlying AI engine ingests real-time behavioral streams and systemic infrastructure data via multiple diagnostic inputs:

- Transaction telemetry payloads (financial values, velocity metrics, account age, and type classifications)
- Geospatial IP mappings and network routing hops to identify anomalous regional variations
- Hardware profiling signatures to verify authentic device environments and cookie histories
- Behavioral dynamics software to flag anomalous tap patterns, scroll rates, or login habits
- Relational core banking ledgers to keep track of historical spending baselines and system error logs

Task 2: Environment Classification

- **Partially Observable:** The analytical engine cannot fully monitor everything, since a bad actor's real intentions, compromised access keys, or hidden coordination tools operate completely outside bank firewalls.
- **Multi-agent:** Millions of cardholders interact concurrently with competing cyber-criminals, where every individual entity shifts the pattern of overall system traffic.
- **Stochastic:** System states are not perfectly deterministic, given that data packets may become corrupted and individual user actions remain highly variable.
- **Sequential:** Actions are deeply interconnected over time; a decision to delay a single payment changes credit risk scores and determines the restrictions placed on upcoming transfers.
- **Dynamic:** The ecosystem undergoes continuous changes as new merchant points go online, consumer spending spikes seasonally, and exploit techniques adapt.
- **Continuous:** System inputs such as monetary amounts, computation intervals, and localized probability distributions occupy an infinite range of fractional values.

Task 3: Critical Analysis

1. Biggest Technical Challenge

The primary technical hurdle is partial observation. In web-scale financial pipelines, an AI system never holds complete visibility over the physical person initiating a bank draft. Security credentials could be legally shared or stolen, device identities can be emulated, and human motives remain obscured. Because of this information asymmetry, the fraud model must draw statistical conclusions from fragmented, circumstantial digital signatures, elevating the risk of locking legitimate funds and necessitating advanced behavioral profiling models.

2. Utility Function (Transaction Volume vs. Risk Protection Trade-off)

A simple utility function can be defined as:

$$U = 0.6 \times OperationalEfficiency - 0.4 \times SystemicRisk$$

This formulation allows the system to establish an optimal baseline balancing rapid volume clearance against potential financial loss. High efficiency maintains processing speeds and platform utility, whereas systemic risk directly degrades banking infrastructure trust.

If the efficiency coefficient is doubled, the system heavily shifts its priority toward unobstructed transaction clearances, bypassing comprehensive evaluation steps. While this boosts transactional revenue short term, it inherently exposes the network to deep financial vulnerabilities and compliance penalties over time.